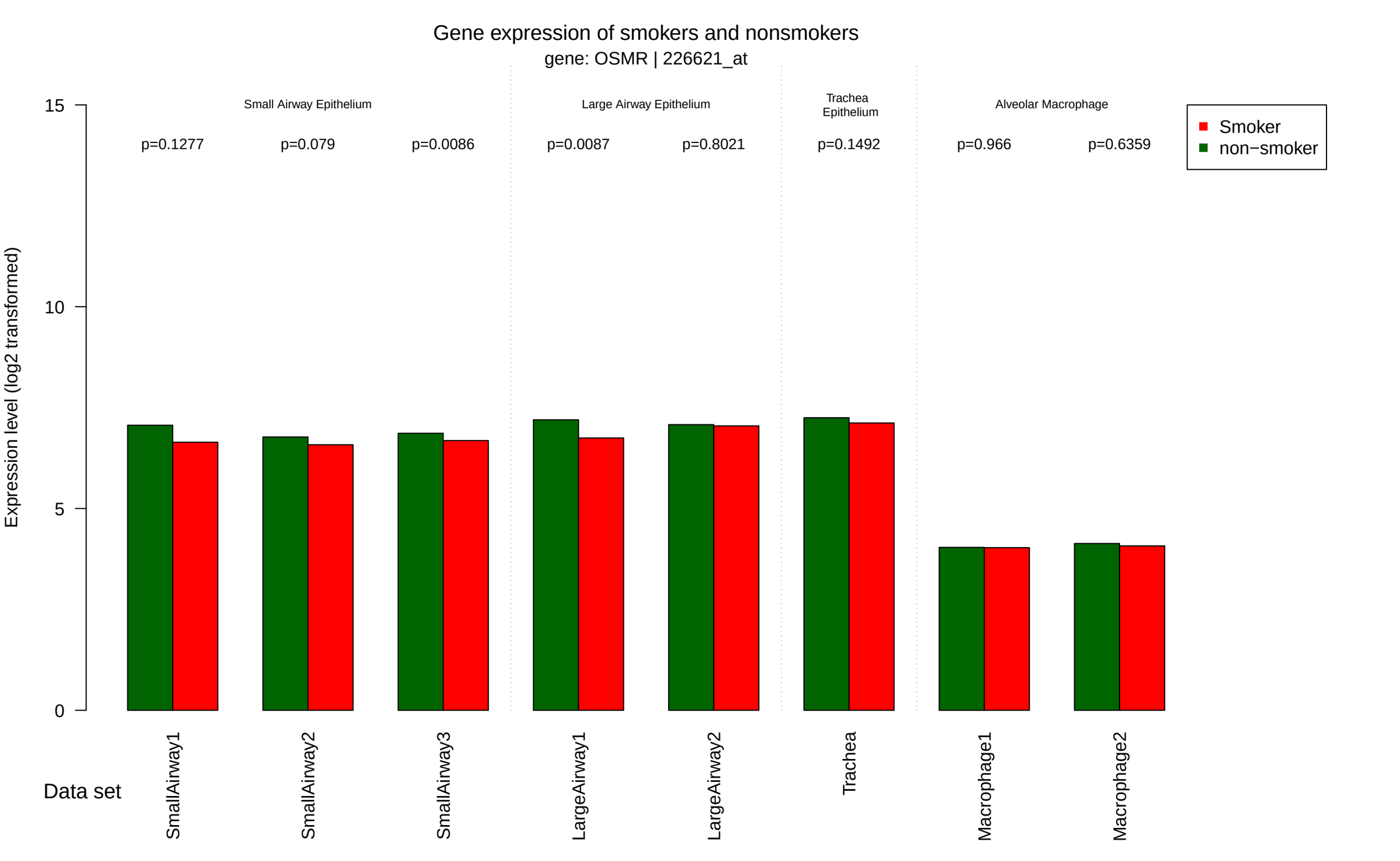
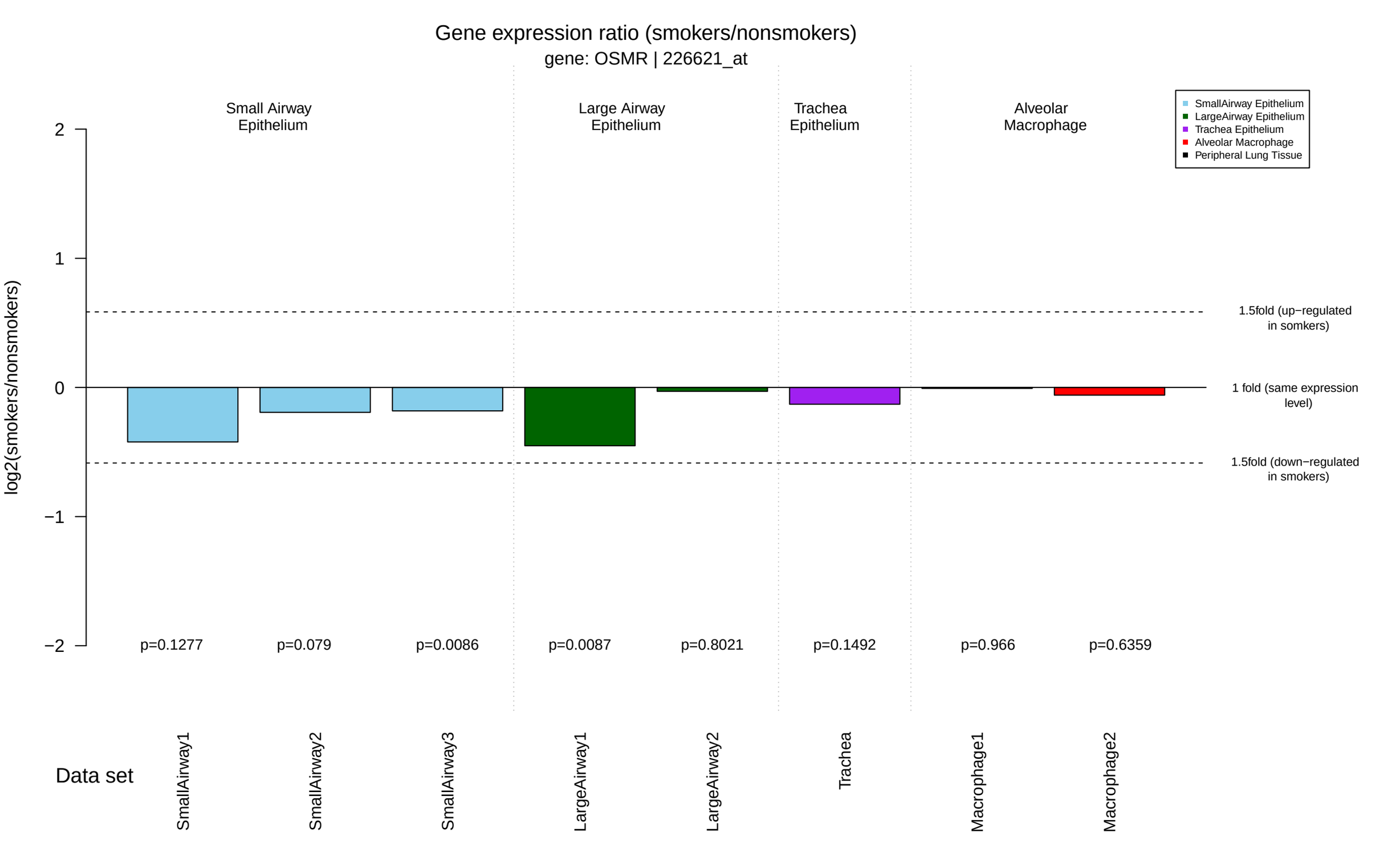
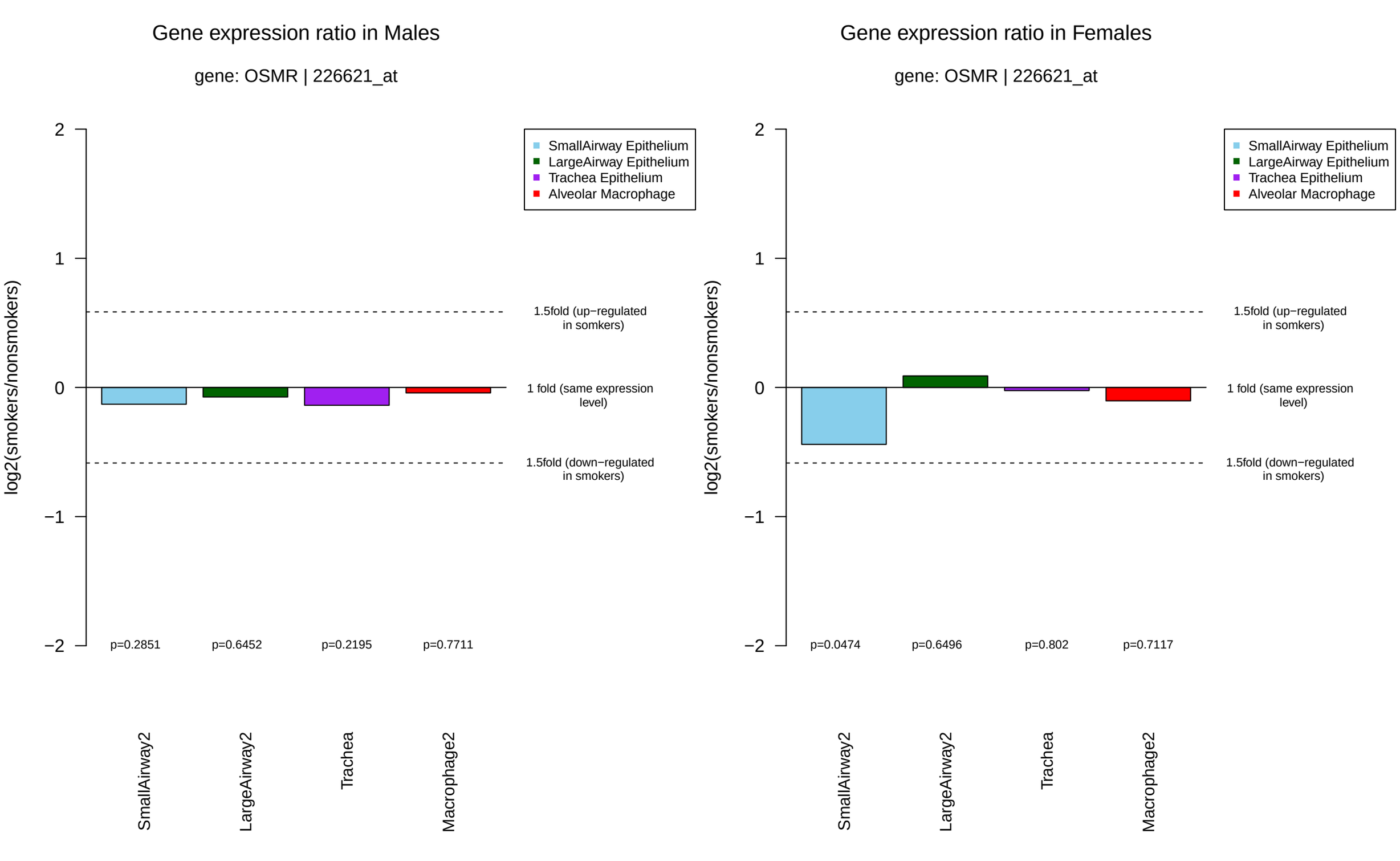
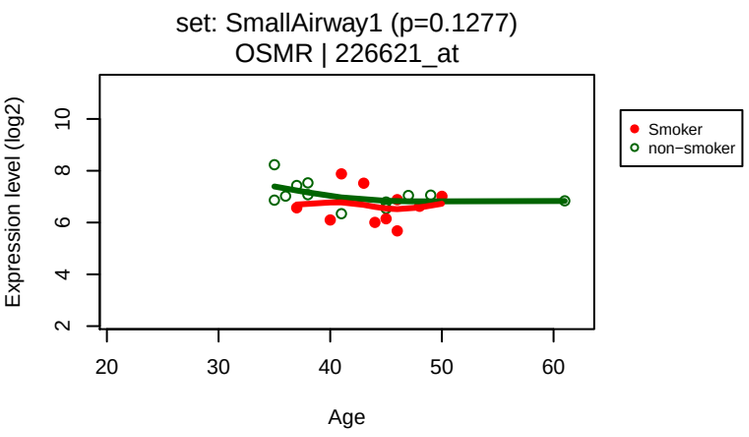
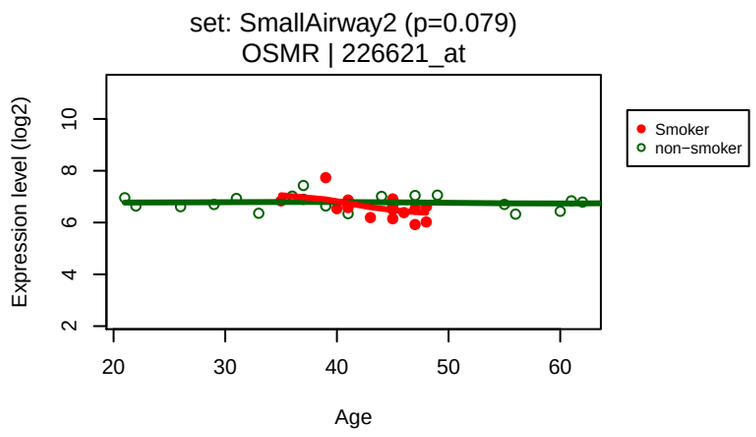
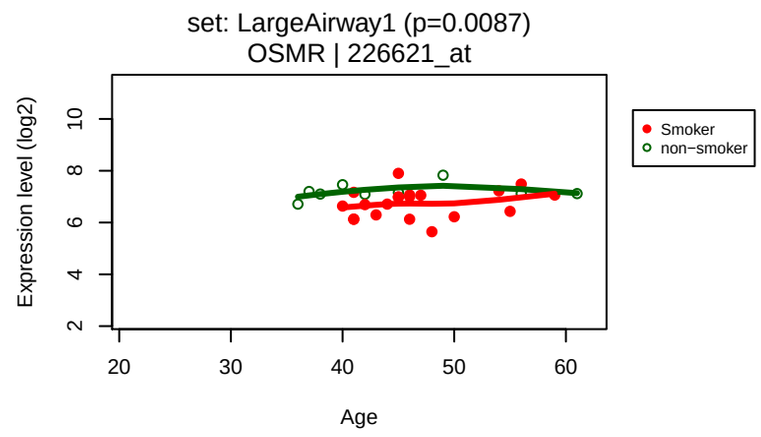

Gene expression of smokers and nonsmokers
gene: OSMR | 226621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1277
p=0.079
p=0.0086
p=0.0087
p=0.8021
p=0.1492
p=0.966
p=0.6359
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

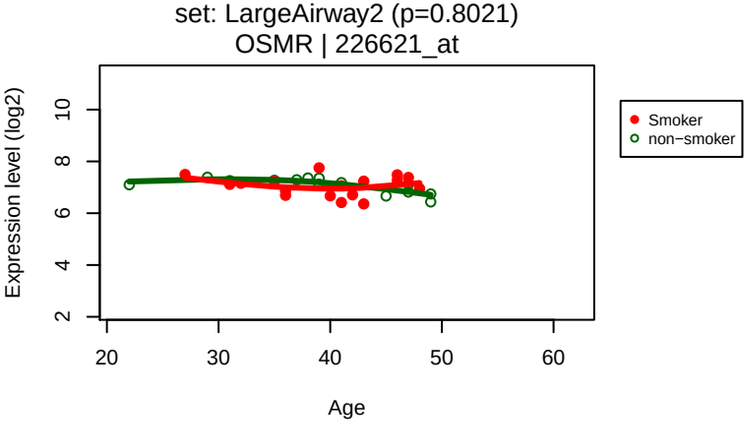

Gene expression ratio (smokers/nonsmokers)
gene: OSMR | 226621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1277
p=0.079
p=0.0086
p=0.0087
p=0.8021
p=0.1492
p=0.966
p=0.6359
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

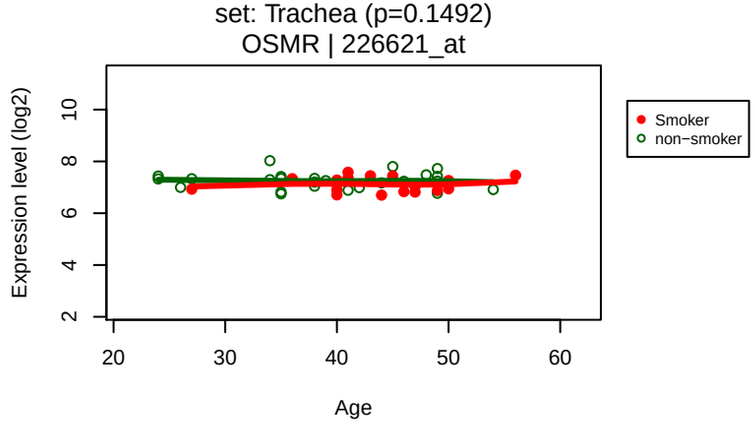

Gene expression ratio in Males
gene: OSMR | 226621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2851
p=0.6452
p=0.2195
p=0.7711
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSMR | 226621_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0474
p=0.6496
p=0.802
p=0.7117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

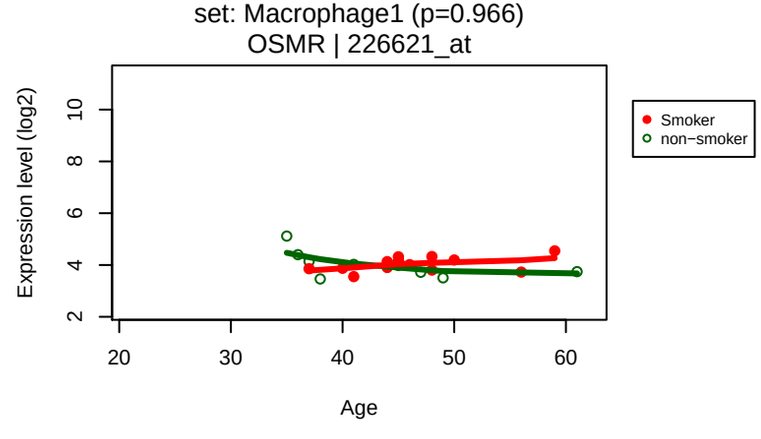

set: SmallAirway1 (p=0.1277)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker

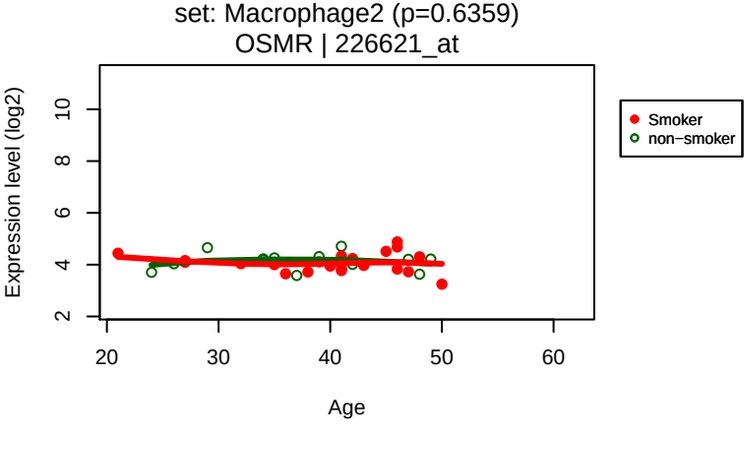

set: SmallAirway2 (p=0.079)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker

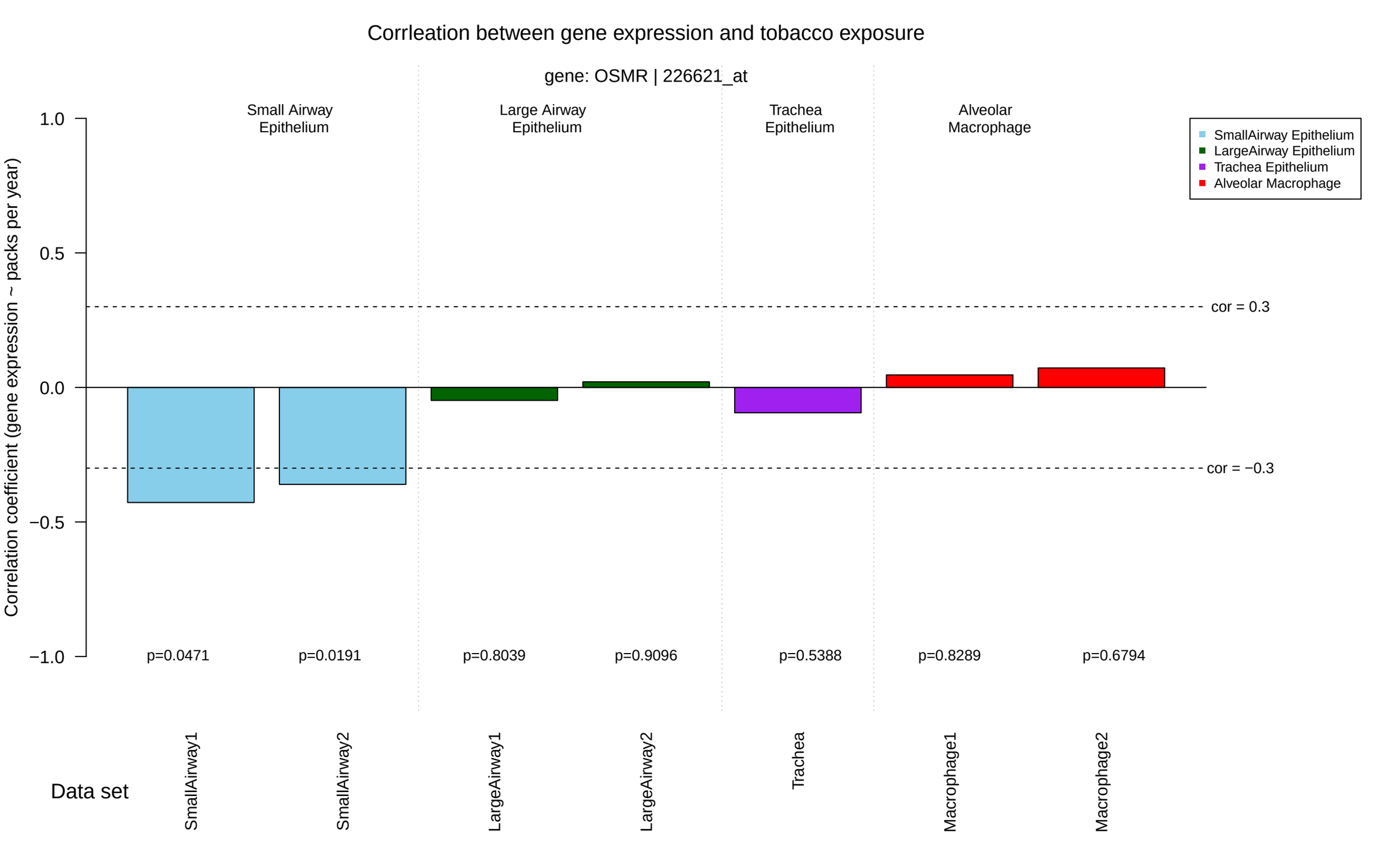

set: LargeAirway1 (p=0.0087)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8021)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1492)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.966)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6359)
OSMR | 226621_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSMR | 226621_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0471
p=0.0191
p=0.8039
p=0.9096
p=0.5388
p=0.8289
p=0.6794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2